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Cloud-Based ITSM Ticket Classification Platform Using Fine-Tuned Transformer Models

Proposal for a project to fulfill

Professional master's degree in Cloud Computing Networks

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Abstract

IT Service Management (ITSM) is a core component of modern enterprise IT operations, requiring accurate classification of large volumes of support tickets by category, subcategory, and priority to ensure proper routing and timely resolution. Manual classification is labor-intensive, error-prone, and can introduce significant delays into service delivery workflows. This challenge is especially evident in Arabic-speaking environments, where there is a notable shortage of Arabic ITSM training data and domain-adapted NLP tools, particularly for colloquial varieties such as Egyptian Arabic.

This project proposes to develop a cloud-based ITSM ticket classification platform with a primary focus on Arabic, with particular attention to Egyptian colloquial usage where feasible. The platform will rely on modern natural language processing techniques and machine learning models suitable for text classification, deployed as scalable cloud services. The system will aim to support multi-label classification (e.g., category, subcategory, and priority) while remaining flexible regarding the exact choice of architectures and tools to allow adaptation as the project progresses.

The project will explore multiple options for obtaining suitable Arabic ITSM-like text, including the potential use of publicly available datasets, and controlled data augmentation techniques. Similarly, different families of machine learning models (such as traditional algorithms, neural networks, and transformer-based approaches) may be evaluated to identify practical trade-offs between performance, complexity, and deployment cost.

At the end of the project, the goal is to deliver a well-documented prototype platform that demonstrates automated ticket classification in Arabic, along with an evidence-based evaluation of its performance under realistic conditions. This evaluation is expected to include standard classification metrics, basic latency and resource usage observations, and qualitative insights into how the system behaves with real or realistic ITSM text. A simple web-based interface will allow users to submit tickets and view classification outputs and associated confidence indications, providing a tangible demonstration of the concept.

By focusing on Arabic ITSM ticket classification and its deployment in a cloud environment, this project aims to contribute to both the practical needs of organizations seeking to improve their IT service workflows and the broader research effort in Arabic natural language processing. The final outcomes are expected to include an initial implementation of the platform, documented experimental results, and a discussion of the strengths, limitations, and potential future extensions.